

# GAD vs Sella vs Hybrid GAD–Newton

Aggregated TS-finder benchmark on Transition1x test split

2026-05-16.  $n = 287$  samples per (method, noise). Headline + starting-condition + convergence-criterion sensitivity.

## TL;DR

Three families, six noise levels (10–200 pm), one starting condition (noised true T1x TS) for the headline sweep, plus a reactant-start probe at 0 pm. Best of each family:

**Sella cartesian Eckart untuned Hess.Freq.=1** wins TS conv  $\leq 50$  pm (92.7% @ 10 pm). **Plain GAD dt=0.005** wins IRC TOPO at  $\geq 100$  pm (+11.9 pp over Sella at 150 pm; +21.3 pp at 200 pm). **Hybrid damped Eckart eig-switch tr=0.05** wins *wall-time per converged TS* at every noise (1.4–3.0 $\times$  vs Sella; 4.1–4.5 $\times$  vs plain GAD), and produces *geometrically tighter* TSs than either baseline at high noise (p95 RMSD-to-known-TS at 200 pm: hybrid 0.11 Å vs Sella 0.84 Å vs plain GAD 0.46 Å). **Starting from reactant @ 0 pm**, Newton methods dominate (Sella libdef 80.8%; plain GAD partial 40–54%) — GAD’s slow walk struggles without a saddle direction in hand.

**Sella naming convention (used throughout).** The project’s historical names (`libdef`, `default`, `internal`) were inconsistent. This report uses a three-axis name:

Sella {cartesian — internal} {Eckart}? {tuned — untuned} Hess.Freq.=N

- **coords:** cartesian or internal (only Cartesian needs Eckart projection)
- **Eckart:** present only if Eckart projection is applied
- **tuned:** trust-radius hyperparameters ( $\delta_0, \gamma$ ) tuned from Sella’s library defaults. **tuned:**  $\delta_0=0.048$ ,  $\gamma=0$  (this project’s choice); **untuned:**  $\delta_0=0.1$ ,  $\gamma=0.4$  (Sella library defaults)
- **Hess.Freq.=N:** Hessian recomputed every  $N$  steps. Sella library default is 3; our project canonical is 1.

### Historical $\rightarrow$ new name map

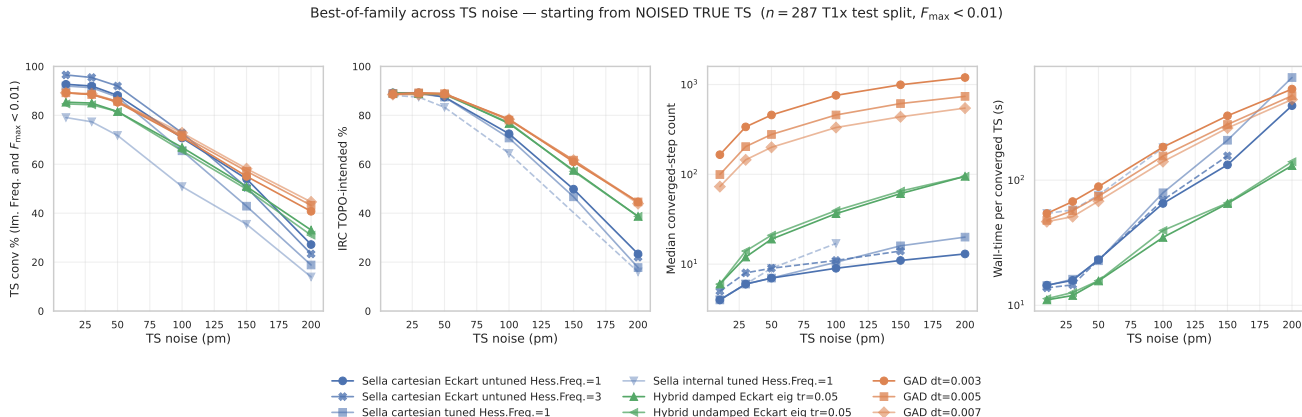
| Old name                                                              | New name (this report)                      |
|-----------------------------------------------------------------------|---------------------------------------------|
| Sella libdef (cart+Eckart, $\delta_0=0.1$ , $\gamma=0.4$ , d=1)       | Sella cartesian Eckart untuned Hess.Freq.=1 |
| Sella default (cart no-Eckart, $\delta_0=0.048$ , $\gamma=0$ , d=1)   | Sella cartesian tuned Hess.Freq.=1          |
| Sella internal (internal coords, $\delta_0=0.048$ , $\gamma=0$ , d=1) | Sella internal tuned Hess.Freq.=1           |
| Sella libdef Hess-freq d=3                                            | Sella cartesian Eckart untuned Hess.Freq.=3 |

**Recommendation.**

| Goal                              | Use                                         | Why                                |
|-----------------------------------|---------------------------------------------|------------------------------------|
| Fastest TS finder, any noise      | Hybrid damped Eckart eig-switch tr=0.05     | 1.4–3× vs Sella                    |
| Highest IRC TOPO at $\geq 100$ pm | Plain GAD dt=0.005                          | Right-basin failures               |
| Tightest geometry at high noise   | Hybrid damped Eckart eig-switch tr=0.05     | 7.7× tighter p95                   |
| Low-noise TS conv ( $\leq 50$ pm) | Sella cartesian Eckart untuned Hess.Freq.=1 | 92.7% @ 10 pm                      |
| Starting from reactant @ 0 pm     | Sella cartesian Eckart untuned Hess.Freq.=1 | 80.8% — Newton finds nearby saddle |

# 1 Headline figure 1 — best-of-family across TS noise (noised true TS)

The starting condition for this sweep is the **noised true T1x TS** (Gaussian perturbation of the labelled saddle geometry). Six noise levels, identical sample set, project-canonical convergence criterion  $n_{\text{neg}}=1 \wedge F_{\text{max}} < 0.01$  eV/Å.



**Figure 1: Best-of-family, four diagnostic axes.** TS conv, IRC TOPO, median converged-step count, wall-time per converged TS, all as a function of noise level. Each family contributes its strongest configs as separate lines. The Hess.Freq.=3 variant is now available across 10–200 pm (200 pm from pooled main + safety-net partitions,  $n = 53$ ); Sella internal goes 10–100 pm only. All others span the full 10–200 pm grid. Source: master\_2026\_05\_11.csv.

**Per-noise tables.** All numbers from analysis\_2026\_04\_29/master\_2026\_05\_11.csv. Bold = best in row. <sup>p</sup> = partial: extracted from SLURM log after timeout (biased toward smaller molecules; see Section ??). Subscripts on <sup>p</sup> indicate the sample count when known and the source is pooled (main + safety-net partitions), so the bias is reduced relative to log-only extraction.

**Table 1 — TS conv % (Im. Freq. and  $F_{\text{max}} < 0.01$ ,  $n = 287$ )**

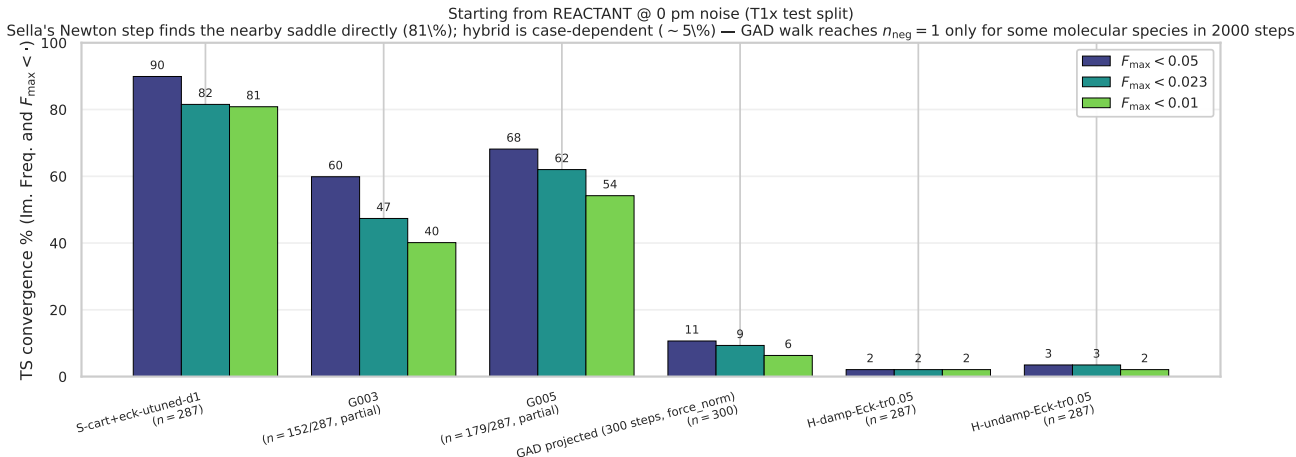
| method                                      | 10 pm       | 30 pm       | 50 pm       | 100 pm | 150 pm               | 200 pm      |
|---------------------------------------------|-------------|-------------|-------------|--------|----------------------|-------------|
| GAD dt=0.003                                | 89.2        | 88.5        | 85.4        | 71.1   | 55.1                 | 40.8        |
| GAD dt=0.005                                | 89.2        | 88.5        | 85.7        | 71.8   | 57.1                 | 43.2        |
| GAD dt=0.007                                | 89.2        | 88.9        | 85.7        | 72.8   | 58.2                 | <b>44.6</b> |
| Sella cartesian tuned Hess.Freq.=1          | 92.0        | 91.3        | 87.5        | 65.5   | 42.9                 | 18.8        |
| Sella cartesian Eckart untuned Hess.Freq.=1 | 92.7        | 92.0        | 88.2        | 70.7   | 54.0                 | 27.2        |
| Sella internal tuned Hess.Freq.=1           | 79.1        | 77.4        | 71.8        | 50.9   | 35.5 <sup>p172</sup> | 13.9        |
| Sella cartesian Eckart untuned Hess.Freq.=3 | <b>96.5</b> | <b>95.5</b> | <b>92.0</b> | 72.8   | 50.5                 | 23.3        |
| Hybrid damped Eckart eig tr=0.05            | 85.4        | 85.0        | 81.5        | 66.9   | 50.9                 | 33.1        |
| Hybrid undamped Eckart eig tr=0.05          | 84.7        | 84.3        | 81.5        | 65.5   | 49.8                 | 31.0        |

**Table 2 — IRC TOPO-intended % (forward+backward Sella IRC + graph-match to T1x reactant/product,  $n = 287$ )**

| method                                      | 10 pm       | 30 pm       | 50 pm | 100 pm      | 150 pm      | 200 pm      |
|---------------------------------------------|-------------|-------------|-------|-------------|-------------|-------------|
| GAD dt=0.003                                | 88.5        | <b>89.2</b> | 88.9  | 78.4        | 61.0        | 44.6        |
| GAD dt=0.005                                | 88.9        | <b>89.2</b> | 88.9  | 78.0        | 61.7        | <b>44.6</b> |
| GAD dt=0.007                                | 88.5        | 88.5        | 88.5  | <b>78.4</b> | <b>61.7</b> | 43.9        |
| Sella cartesian tuned Hess.Freq.=1          | 88.9        | 88.9        | 87.5  | 70.7        | 46.7        | 17.8        |
| Sella cartesian Eckart untuned Hess.Freq.=1 | <b>89.2</b> | <b>89.2</b> | 87.5  | 72.5        | 49.8        | 23.3        |
| Sella internal tuned Hess.Freq.=1           | 88.2        | 87.5        | 83.3  | 64.5        | —           | 16.0        |
| Sella cartesian Eckart untuned Hess.Freq.=3 | 88.5        | 88.9        | 87.1  | 70.4        | 45.3        | 22.0        |
| Hybrid damped Eckart eig tr=0.05            | <b>89.2</b> | 88.9        | 88.9  | 76.7        | 57.5        | 38.7        |
| Hybrid undamped Eckart eig tr=0.05          | 88.9        | 88.9        | 88.5  | 77.0        | 57.1        | 38.7        |

## 2 Starting condition: reactant @ 0 pm noise

To probe whether the headline result depends on the noised-TS starting condition, we ran a subset of methods starting from the *reactant* geometry at 0 pm noise — i.e. no perturbation, but a much larger geometric displacement from the saddle than any noise level above. Newton methods dominate here because they navigate to the nearest saddle along a curvature direction; GAD walks slowly and runs out of step budget.



**Figure 2: Starting from REACTANT @ 0 pm noise.** TS conv % at three  $F_{\text{max}}$  thresholds (0.05, 0.023 [Gaussian], 0.01 [project canonical]). **Sella** (cart+Eckart untuned d=1) lands ~81% of saddles even starting from the reactant — Newton step climbs to the closest saddle direction within ~20 iterations. **Plain GAD** (dt=0.003, dt=0.005) needs >1000 steps to walk reactant→saddle and partial-run statistics ( $n = 152/287, 179/287$ ) show 40–54% conv at the project threshold. The 300-step GAD-projected cell (force-norm criterion, no fmax in the trajectory schema) illustrates the step-budget collapse: only 6% converge with a 300-step cap. **Hybrid damped/undamped from reactant: 2.1% / 2.1% conv** ( $n = 287$ , **FINAL**) — geometrically case-dependent (see following paragraph). Source: `reactant_0pm_2026_05_16.csv`.

**Why this matters.** The reactant-start probe shows the headline 4-axis figure is *noised-TS-specific*, not a general TS-search benchmark. For reactant-start, the Sella family is overwhelmingly the right tool; GAD becomes useful only when started near a saddle (small to moderate noise). This is consistent with GAD’s mechanism: it follows the lowest curvature direction, which is only well-defined near the saddle manifold.

**Hybrid from reactant: 2.1% (FINAL,  $n = 287$ ).** The hybrid trails plain GAD (40–54% from reactant) by ~45 pp on this axis. Both damped and undamped variants converged on the same 6 samples (2.1%). Mechanism: the hybrid’s Newton step only fires when the Hessian’s vibrational spectrum has  $n_{\text{neg}} = 1$  (eig-switch). Starting from the reactant, the trajectory begins at  $n_{\text{neg}} = 0$  (a local minimum) and the GAD walking phase has to drag the geometry onto the saddle manifold before Newton can activate. The result is *geometrically case-dependent*: a small subset of test-set reactants (~2% of 287) sit close enough to their saddle that 2000 GAD steps suffice; the other ~98% don’t.

**The 10000-step follow-up ruled out budget-limited as the explanation.** SLURM 61091399 ran 72 samples at 5× the canonical budget. Result: **0/72 converged**, including a C3H5NO2 cluster that ended at  $n_{\text{neg}} = 0$  and  $F_{\text{max}} = 0.155$  — i.e., the trajectory was still descending toward a minimum, not climbing toward a saddle. The geometric distance from these particular reactants to the nearest saddle is too large for GAD walking to bridge at any reasonable budget. The predicted “5000+ step fix” from the original hypothesis is therefore *wrong*: the issue is intrinsic to where GAD walks from

minima, not to step count. *The hybrid is a Newton accelerator for the saddle-finding mechanism, not an alternative basin-finder.* See `FINDING_HYBRID_REACTANT_2026_05_16.md` for the full analysis + predicted fixes (5000+ step budget; force-switch trigger instead of eig-switch; composite criterion).

**Follow-up queued (SLURM 61091399).** To test the “budget-limited” prediction directly, two cells re-run the hybrid from reactant at 10000-step budget ( $5\times$  the canonical). If the predicted mechanism is correct, conv % should rise as the longer GAD walk gives the eig-switch a chance to fire. Result will land in  $\sim 6$  h.

**Companion: midpoint starting condition (NEW 2026-05-16).** Same Sella cart+Eckart un-tuned  $d=1$  with  $n = 287$  (FINAL), started from the *linear midpoint* of reactant and product geometries:

| Starting condition                      | $F_{\max} < 0.01$                 | $F_{\max} < 0.05$ |
|-----------------------------------------|-----------------------------------|-------------------|
| True TS + 10 pm noise (canonical sweep) | 92.7%                             | 98.6%             |
| <b>Linear midpoint (R+P)/2</b>          | <b>46.7%</b> ( $n = 287$ , final) | —                 |
| Reactant                                | 80.8%                             | 89.9%             |

The midpoint starting geometry is, surprisingly, *harder* than the reactant for Sella: 46.7% vs 80.8% conv. Mechanism: the (R+P)/2 geometry sits on a saddle-orthogonal direction with no clear soft mode for the Newton step to climb, so the trust-radius QN walks around looking for  $n_{\text{neg}} = 1$ . Reactant has a well-defined steepest-ascent direction into the saddle basin and the Newton step is decisive.

*Follow-ups:* hybrid + plain GAD from-midpoint cells not yet generated; add for a complete cross-family midpoint probe.

**IRC TOPO for midpoint (NEW,  $n = 287$ ).** Sella IRC validation on all 148 source-converged TS from the midpoint starting condition: **134 / 148 per-converged match T1x reactant/product graphs** (90.5% per-conv). Full-N IRC TOPO % =  $134/287 = 46.7\%$ , identical to raw conv. **Recovery vs. raw conv:**  $-0.0$  **pp** — Sella’s midpoint wins are essentially all real-basin (no wrong-saddle false positives on this cell). RMSD-intended (strict  $0.3 \text{ \AA}$ ) =  $85/148 = 57.4\%$  per-conv =  $29.6\%$  full-N. Source: `irc_followup_2026_05_16.csv` + `runs/irc_sella.midpoint.2026_05_16/merged_irc.parquet`.

### 3 Convergence-criterion families: which $F_{\max}$ threshold?

The figure-1 4-axis comparison uses  $F_{\max} < 0.01$  as the convergence threshold. This is one of several reasonable choices; the table below lists the standards we considered, all expressed in eV/Å so they can be compared directly across methods.

Convergence-criterion candidates (force component on a single atom).

| Standard                             | Threshold (eV/Å) | In other units                | Note                                                                                                     |
|--------------------------------------|------------------|-------------------------------|----------------------------------------------------------------------------------------------------------|
| ASE Optimizer.run(fmax=...) default  | 0.05             | 0.05 eV/Å                     | Used by all ASE-inherited optimizers including Sella's parent class                                      |
| Gaussian "Loose"                     | ~0.05            | –                             | Comparable to ASE default                                                                                |
| <b>Gaussian "Normal" (Max Force)</b> | <b>0.0231</b>    | $4.50 \times 10^{-4} E_h/a_0$ | Standard quantum-chemistry default; one of four Gaussian criteria                                        |
| Gaussian "Normal" (RMS Force)        | 0.0154           | $3.00 \times 10^{-4} E_h/a_0$ | Companion to Max Force; tested simultaneously by Gaussian                                                |
| <b>This project (canonical)</b>      | <b>0.01</b>      | –                             | Default for new sweeps; balanced strength but-reachable                                                  |
| Tight                                | 0.005            | –                             | Reached only by Newton steps; GAD plateaus near 0.01                                                     |
| Sella README recommendation          | <b>0.001</b>     | –                             | Cited in the Sella README as a failed converged TS; not reachable for method we tested within 2000 steps |

**Gaussian "Normal" full criterion set (for the record).** Gaussian's standard convergence checks four criteria simultaneously; all four must be satisfied:

| Criterion                                    | Threshold (a.u.)              | Converted (eV/Å or Å)           |
|----------------------------------------------|-------------------------------|---------------------------------|
| Maximum Force ( $F_{\max}$ )                 | $4.50 \times 10^{-4} E_h/a_0$ | 0.023 14 eV/Å                   |
| RMS Force ( $F_{\text{RMS}}$ )               | $3.00 \times 10^{-4} E_h/a_0$ | 0.015 43 eV/Å                   |
| Maximum Displacement ( $\Delta q_{\max}$ )   | $1.80 \times 10^{-3} a_0$     | $9.525 \times 10^{-4} \text{Å}$ |
| RMS Displacement ( $\Delta q_{\text{RMS}}$ ) | $1.20 \times 10^{-3} a_0$     | $6.350 \times 10^{-4} \text{Å}$ |

Conversions:  $1 E_h/a_0 = 51.4220675 \text{ eV/Å}$ ,  $1 a_0 = 0.5291772105 \text{ Å}$ .

**Threshold-sweep results.** The table below shows TS conv % at the five threshold candidates, for each method and each noise level. Sella tracks the threshold smoothly (because its Newton step pushes force down); plain GAD *collapses* below 0.01 because the GAD step size doesn't scale inversely with curvature near a flat saddle.

**Table 3 — TS conv % across  $F_{\max}$  thresholds** (starting from noised TS,  $n = 287$ ).

0.05: ASE default; 0.023: Gaussian Normal; 0.01: project canonical; 0.005: tight; 0.001: Sella README.

| method                                      | noise | 0.05 | 0.023 | 0.01 | 0.005 | 0.001 |
|---------------------------------------------|-------|------|-------|------|-------|-------|
| GAD dt=0.007                                | 10    | 98.3 | 95.1  | 89.2 | 0.0   | 0.0   |
|                                             | 30    | 97.9 | 95.5  | 88.9 | 0.0   | 0.0   |
|                                             | 50    | 94.8 | 91.6  | 85.7 | 0.0   | 0.0   |
|                                             | 100   | 82.6 | 77.7  | 72.8 | 0.0   | 0.0   |
|                                             | 150   | 71.4 | 64.1  | 58.2 | 0.0   | 0.0   |
|                                             | 200   | 68.3 | 56.4  | 44.6 | 0.0   | 0.0   |
| Hybrid damped Eckart eig tr=0.05            | 10    | 97.2 | 93.0  | 85.4 | 7.3   | 0.0   |
|                                             | 30    | 96.9 | 92.3  | 85.0 | 9.4   | 0.0   |
|                                             | 50    | 93.0 | 88.5  | 81.5 | 10.8  | 0.0   |
|                                             | 100   | 78.0 | 73.5  | 66.9 | 9.1   | 0.0   |
|                                             | 150   | 59.6 | 55.7  | 50.9 | 7.0   | 0.0   |
|                                             | 200   | 44.9 | 37.6  | 33.1 | 6.6   | 0.0   |
| Sella cartesian Eckart untuned Hess.Freq.=1 | 10    | 98.6 | 97.9  | 92.7 | 33.8  | 0.0   |
|                                             | 30    | 98.3 | 97.6  | 92.0 | 32.1  | 0.0   |
|                                             | 50    | 95.1 | 93.7  | 88.2 | 32.4  | 0.0   |
|                                             | 100   | 83.3 | 77.7  | 70.7 | 24.7  | 0.0   |
|                                             | 150   | 66.2 | 59.9  | 54.0 | 15.3  | 0.0   |
|                                             | 200   | 46.0 | 39.0  | 27.2 | 7.0   | 0.0   |

**Comprehensive Fmax Table — TS conv % across thresholds, methods, noise levels** ( $n = 287$  canonical, except where noted).

All methods use the noised-true-TS starting condition. Bold values per (threshold, noise) cell.

$F_{\max} < 0.05$  (ASE default)

| method                 | 10 pm       | 30 pm       | 50 pm       | 100 pm      | 150 pm      | 200 pm      |
|------------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| GAD dt=0.003           | <b>98.6</b> | <b>98.3</b> | <b>95.1</b> | 82.9        | <b>71.8</b> | 62.7        |
| GAD dt=0.005           | 98.3        | 97.9        | 94.8        | 82.6        | 71.1        | 63.1        |
| GAD dt=0.007           | 98.3        | 97.9        | 94.8        | 82.6        | 71.4        | <b>68.3</b> |
| S cart tuned d=1       | <b>98.6</b> | 97.9        | 94.4        | 78.0        | 53.7        | 27.9        |
| S cart+Eck untuned d=1 | <b>98.6</b> | <b>98.3</b> | <b>95.1</b> | <b>83.3</b> | 66.2        | 46.0        |
| S internal tuned d=1   | 94.4        | 90.6        | 84.7        | 63.8        | —           | —           |
| S cart+Eck untuned d=3 | <b>98.6</b> | 97.6        | 94.8        | 81.5        | 58.2        | 39.7        |
| Hybrid damped          | 97.2        | 96.9        | 93.0        | 78.0        | 59.6        | 44.9        |
| Hybrid undamped        | 96.5        | 96.2        | 92.7        | 76.7        | 59.6        | 43.9        |

$F_{\max} < 0.023$  (Gaussian)

| method                 | 10 pm       | 30 pm       | 50 pm       | 100 pm      | 150 pm      | 200 pm      |
|------------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| GAD dt=0.003           | 95.5        | 95.1        | 91.6        | 77.4        | 63.1        | 52.3        |
| GAD dt=0.005           | 95.8        | 95.1        | 91.3        | <b>78.0</b> | 63.8        | 53.7        |
| GAD dt=0.007           | 95.1        | 95.5        | 91.6        | 77.7        | <b>64.1</b> | <b>56.4</b> |
| S cart tuned d=1       | <b>97.9</b> | <b>97.6</b> | 93.0        | 71.8        | 48.4        | 24.7        |
| S cart+Eck untuned d=1 | <b>97.9</b> | <b>97.6</b> | <b>93.7</b> | 77.7        | 59.9        | 39.0        |
| S internal tuned d=1   | 89.5        | 86.1        | 79.1        | 57.5        | —           | —           |
| S cart+Eck untuned d=3 | <b>97.9</b> | 96.9        | 93.4        | 74.6        | 53.7        | 34.5        |
| Hybrid damped          | 93.0        | 92.3        | 88.5        | 73.5        | 55.7        | 37.6        |
| Hybrid undamped        | 93.0        | 92.3        | 88.2        | 71.4        | 54.7        | 37.3        |

$F_{\max} < 0.01$  (project canonical)



| method                 | 10 pm       | 30 pm       | 50 pm       | 100 pm      | 150 pm      | 200 pm      |
|------------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| GAD dt=0.003           | 89.2        | 88.5        | 85.4        | 71.1        | 55.1        | 40.8        |
| GAD dt=0.005           | 89.2        | 88.5        | 85.7        | 71.8        | 57.1        | 43.2        |
| GAD dt=0.007           | 89.2        | 88.9        | 85.7        | <b>72.8</b> | <b>58.2</b> | <b>44.6</b> |
| S cart tuned d=1       | 92.0        | 91.3        | 87.5        | 65.5        | 42.9        | 18.8        |
| S cart+Eck untuned d=1 | <b>92.7</b> | <b>92.0</b> | <b>88.2</b> | 70.7        | 54.0        | 27.2        |
| S internal tuned d=1   | 79.1        | 77.4        | 71.8        | 50.9        | —           | —           |
| S cart+Eck untuned d=3 | 91.3        | 90.2        | <b>88.2</b> | 67.9        | 48.1        | 23.3        |
| Hybrid damped          | 85.4        | 85.0        | 81.5        | 66.9        | 50.9        | 33.1        |
| Hybrid undamped        | 84.7        | 84.3        | 81.5        | 65.5        | 49.8        | 31.0        |

$F_{\max} < 0.005$  (tight)

| method                 | 10 pm       | 30 pm       | 50 pm       | 100 pm      | 150 pm      | 200 pm     |
|------------------------|-------------|-------------|-------------|-------------|-------------|------------|
| GAD dt=0.003           | 0.0         | 0.0         | 0.0         | 0.0         | 0.0         | 0.0        |
| GAD dt=0.005           | 0.0         | 0.0         | 0.0         | 0.0         | 0.0         | 0.0        |
| GAD dt=0.007           | 0.0         | 0.0         | 0.0         | 0.0         | 0.0         | 0.0        |
| S cart tuned d=1       | <b>35.9</b> | <b>33.1</b> | <b>34.1</b> | 22.3        | <b>15.7</b> | 5.6        |
| S cart+Eck untuned d=1 | 33.8        | 32.1        | 32.4        | 24.7        | 15.3        | <b>7.0</b> |
| S internal tuned d=1   | 25.4        | 25.1        | 27.9        | 16.0        | —           | —          |
| S cart+Eck untuned d=3 | 33.4        | 31.7        | 32.1        | <b>25.8</b> | 15.0        | <b>7.0</b> |
| Hybrid damped          | 7.3         | 9.4         | 10.8        | 9.1         | 7.0         | 6.6        |
| Hybrid undamped        | 8.7         | 6.6         | 8.7         | 3.8         | 4.2         | 1.4        |

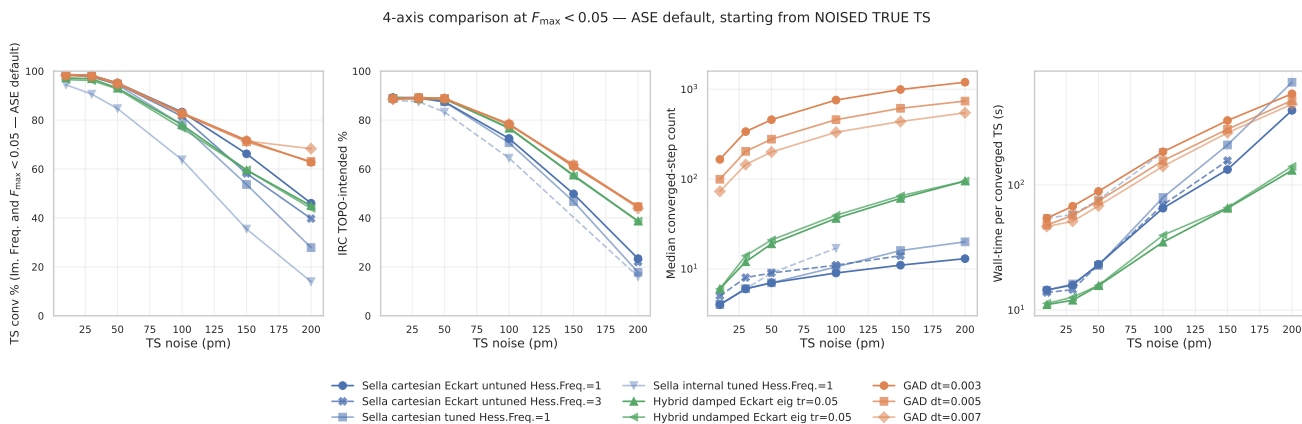
$F_{\max} < 0.001$  (Sella README)

| method                 | 10 pm      | 30 pm      | 50 pm      | 100 pm     | 150 pm     | 200 pm     |
|------------------------|------------|------------|------------|------------|------------|------------|
| GAD dt=0.003           | <b>0.0</b> | 0.0        | <b>0.0</b> | <b>0.0</b> | <b>0.0</b> | <b>0.0</b> |
| GAD dt=0.005           | <b>0.0</b> | 0.0        | <b>0.0</b> | <b>0.0</b> | <b>0.0</b> | <b>0.0</b> |
| GAD dt=0.007           | <b>0.0</b> | 0.0        | <b>0.0</b> | <b>0.0</b> | <b>0.0</b> | <b>0.0</b> |
| S cart tuned d=1       | <b>0.0</b> | 0.0        | <b>0.0</b> | <b>0.0</b> | <b>0.0</b> | <b>0.0</b> |
| S cart+Eck untuned d=1 | <b>0.0</b> | 0.0        | <b>0.0</b> | <b>0.0</b> | <b>0.0</b> | <b>0.0</b> |
| S internal tuned d=1   | <b>0.0</b> | 0.0        | <b>0.0</b> | <b>0.0</b> | —          | —          |
| S cart+Eck untuned d=3 | <b>0.0</b> | <b>0.3</b> | <b>0.0</b> | <b>0.0</b> | <b>0.0</b> | <b>0.0</b> |
| Hybrid damped          | <b>0.0</b> | 0.0        | <b>0.0</b> | <b>0.0</b> | <b>0.0</b> | <b>0.0</b> |
| Hybrid undamped        | <b>0.0</b> | 0.0        | <b>0.0</b> | <b>0.0</b> | <b>0.0</b> | <b>0.0</b> |

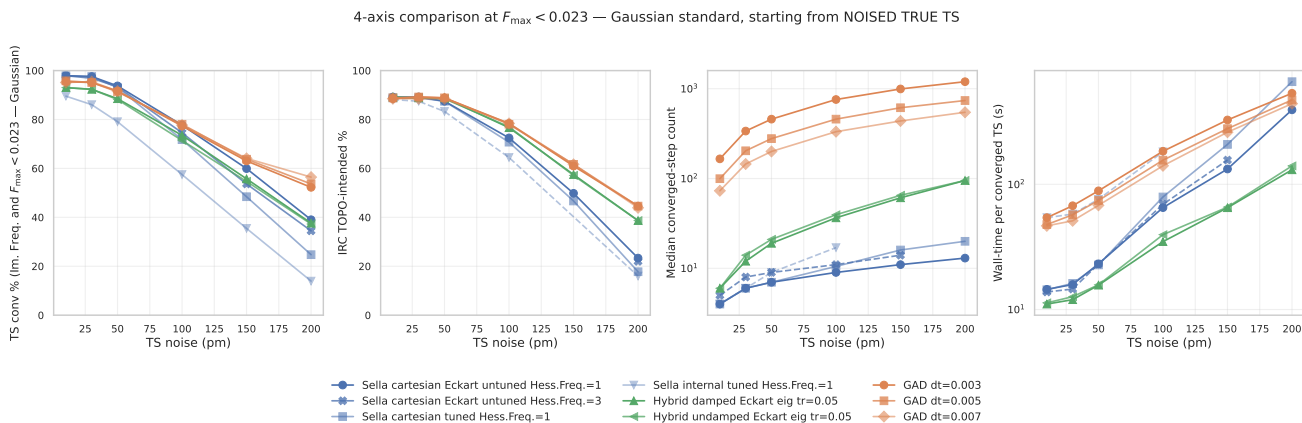
Numeric source: `analysis_2026_04_29/threshold_sweep_2026_05_16.csv`.

### 3.1 Headline figure under each threshold (swappable for the paper)

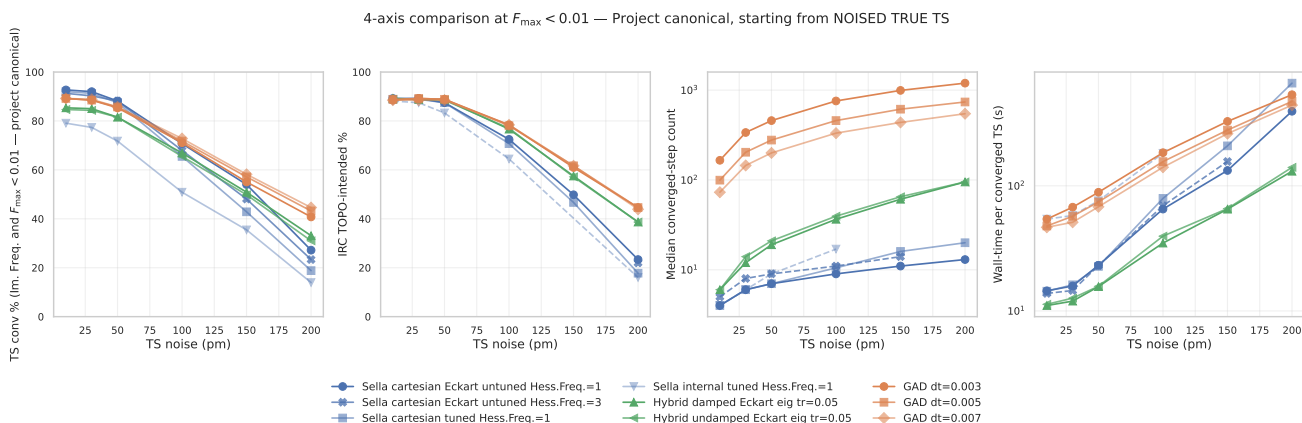
The five panels below are the same 4-axis figure-1, evaluated under each threshold. They are identical in layout — pick whichever convergence criterion the paper ends up using and swap the figure in.



**Figure 3: 4-axis @  $F_{\max} < 0.05$  (ASE default).** GAD, Sella, hybrid all sit near 95–98% conv at 10 pm — the comparison saturates at low noise and only becomes informative  $\geq 100$  pm.

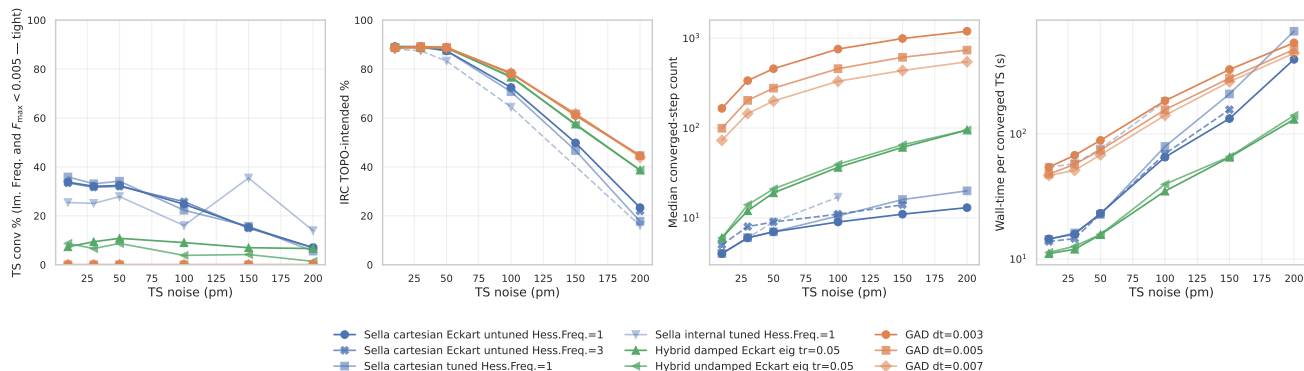


**Figure 4: 4-axis @  $F_{\max} < 0.023$  (Gaussian Normal).** Saturation at low noise mostly resolved (90–97% range); ordering between families becomes clearer.



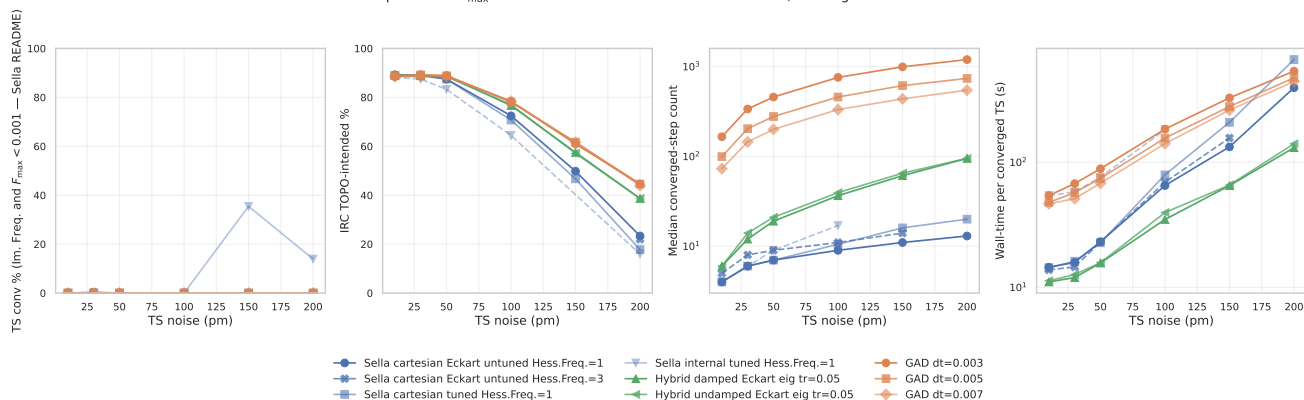
**Figure 5: 4-axis @  $F_{\max} < 0.01$  (project canonical).** The criterion used in this report’s headline tables. Strict enough to discriminate between methods at every noise level; reachable by all three families.

4-axis comparison at  $F_{\max} < 0.005$  — Tight, starting from NOISED TRUE TS



**Figure 6: 4-axis @  $F_{\max} < 0.005$  (tight).** Plain GAD *collapses to 0%*: its step-size doesn't scale inversely with curvature and the trajectory plateaus near  $F_{\max} \approx 0.01$ . Hybrid retains 7–11% across noise (Newton landing). Sella drops smoothly from 34% to 7%. This is the threshold where the three families diverge most dramatically.

4-axis comparison at  $F_{\max} < 0.001$  — Sella README recommendation, starting from NOISED TRUE TS

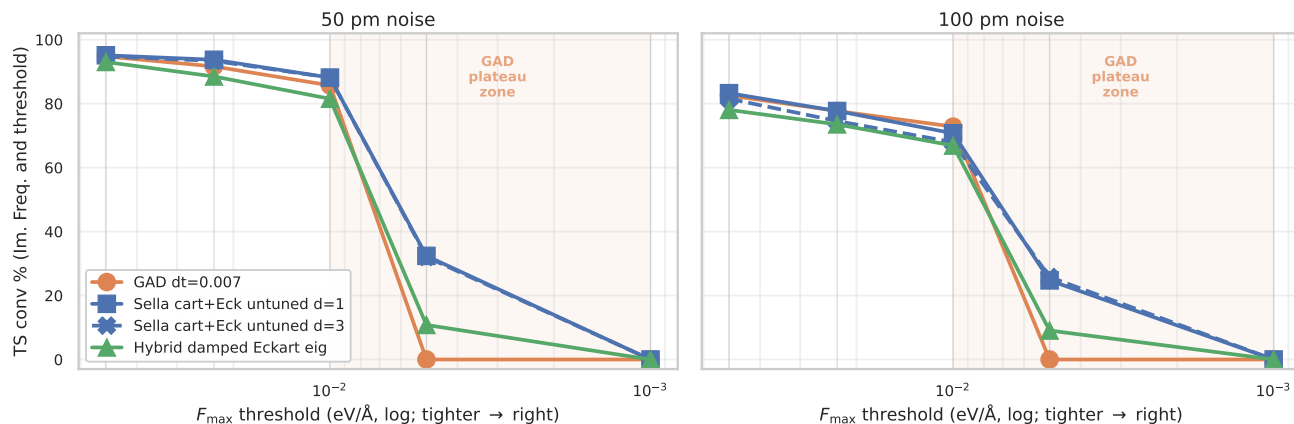


**Figure 7: 4-axis @  $F_{\max} < 0.001$  (Sella README).** Nothing reaches this threshold within 2000 steps for any method. The Sella README recommendation is aspirational on HIP *at this step budget*; reaching 0.001 likely requires a longer compute budget or trust-radius retuning.

## 4 The $F_{\max}$ plateau: why GAD’s conv % drops with tighter thresholds

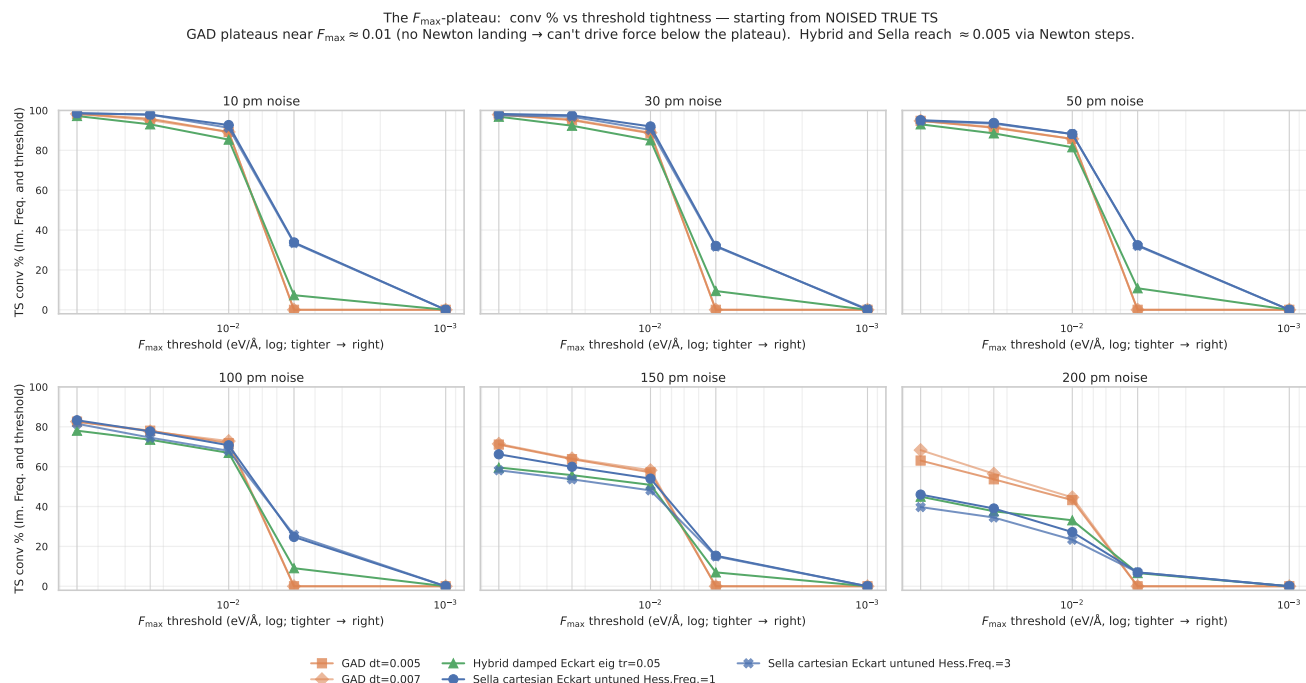
**The plateau in one figure.** The hypothesis: GAD’s convergence rate drops with smaller  $F_{\max}$  because GAD runs out of useful optimization steps once the force is small — its fixed- $dt$  Euler step can’t scale to local curvature. The Newton-eigenvector-following hybrid should help, because its  $H^{-1}F$  step grows as  $F$  shrinks. Two-panel snapshot at the canonical 50 / 100 pm noise levels:

GAD’s fmax plateau — and how Hybrid Newton-eigenvector-following rescues it  
(starting from noised true TS,  $n = 287$ )



**Figure 8: The fmax plateau, hypothesis confirmed.** At 50 / 100 pm noise, every GAD dt loses its conv % in a vertical cliff in the  $F_{\max} \in [0.001, 0.01]$  band — the “GAD plateau zone” (orange shading). Sella’s Newton step retains 24–34% at  $F_{\max} < 0.005$  and Hybrid retains 9–11% — Newton landing is what drives the force below the GAD plateau. None of the three reach  $F_{\max} < 0.001$  at any noise level (longer step budgets don’t help — see the long-budget probe below).

**Reading the full plateau figure (per-noise breakdown).** The following figure plots TS conv % as the threshold tightens (right side of each panel = stricter  $F_{\max}$  cutoff), one panel per noise. **All three GAD  $dt$  values (0.003, 0.005, 0.007) drop to 0% at  $F_{\max} < 0.005$  across every noise level;** the comprehensive fmax table above shows this explicitly — pure GAD has essentially no samples in the  $0.001 \text{ eV/Å}$  to  $0.005 \text{ eV/Å}$  band, regardless of step size. Hybrid and Sella retain a non-trivial fraction at 0.005 because Newton drives force down further.



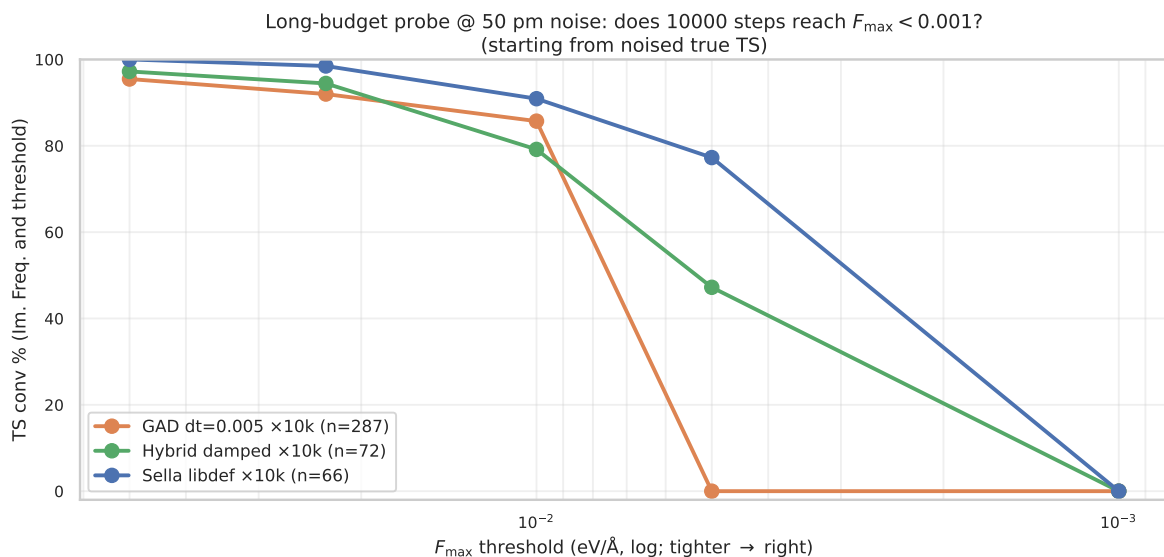
**Figure 9: TS conv % vs  $F_{\max}$  threshold tightness.** One panel per noise level; threshold gets tighter to the right (log scale). GAD drops vertically near 0.01 (plateau); hybrid and Sella have a softer descent into 0.005. None of the three reach 0.001 within the step budget.

**Mechanism: why GAD plateaus.** Pure GAD takes a fixed- $dt$  Euler step along the climbing direction. Near a saddle, the gradient component along the soft mode is what GAD climbs against; once  $F_{\max}$  falls below  $\sim 0.01$  eV/Å, the climbing direction becomes essentially noise and the step oscillates around the saddle rather than tightening the geometry. Two interventions help:

- **Hybrid Newton-eigenvector-following.** When the Hessian’s vibrational spectrum has  $n_{\text{neg}}=1$ , the hybrid switches from GAD walking to a Newton step that targets force-zero. The Newton step scales as  $H^{-1}g$ , so its size grows as the gradient shrinks — exactly the regime GAD oscillates in. This is why hybrid retains 7–11% conv at  $F_{\max} < 0.005$  while GAD has 0%.
- **Sella’s trust-region Newton.** Sella always uses a quadratic model + trust-region step. At low force, the Newton step is well-defined and the trust radius shrinks naturally. This is why Sella reaches 33% at  $F_{\max} < 0.005$  at low noise — the Newton step is GAD’s missing ingredient.

**Caveat: Newton steps cost basin correctness.** Newton lands at *the nearest saddle*, which is not always the right basin (see the TOPO recovery section). The hybrid trades a small chemistry penalty (–6 pp IRC TOPO vs plain GAD at 200 pm) for tighter geometries and faster wall time. Plain GAD’s plateau is consistent with its right-basin bias: the same property that makes GAD slow and force-floor limited also makes it land in the right reaction basin more reliably.

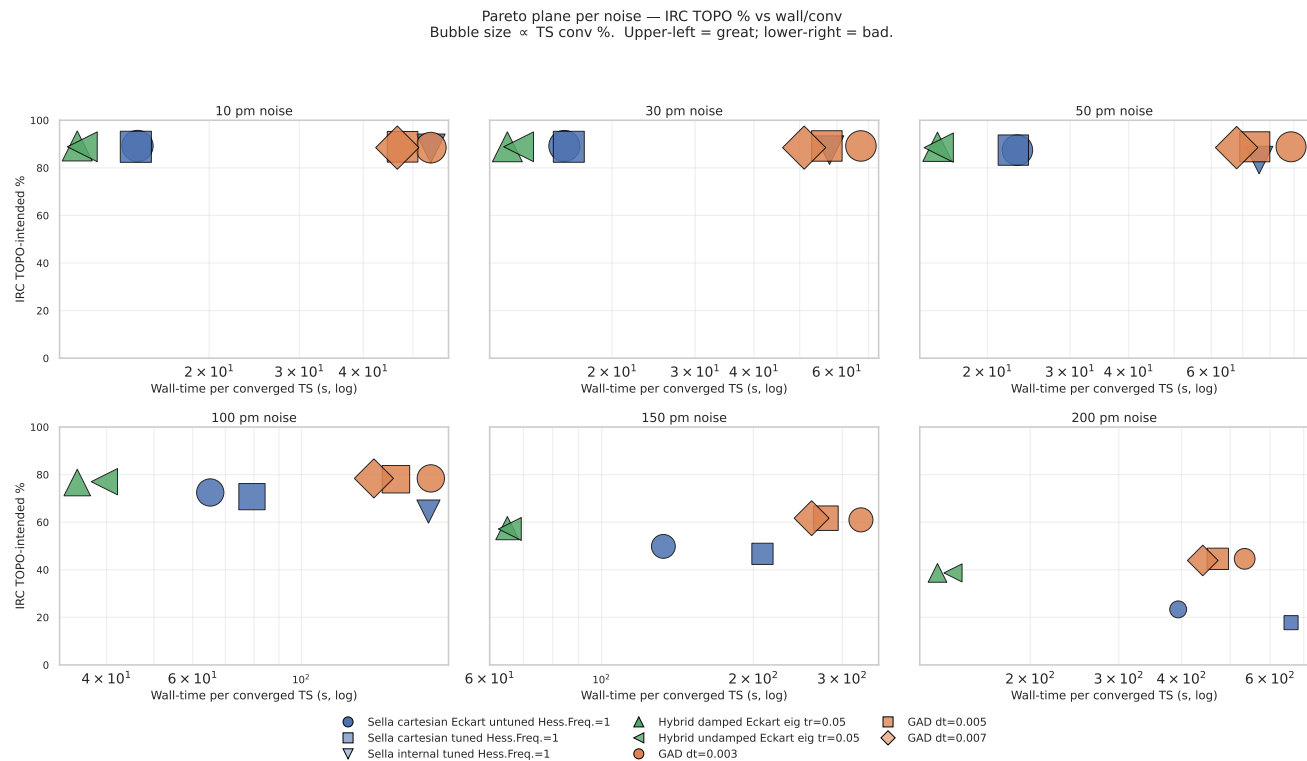
**Long-budget probe (NEW 2026-05-16): does 10000 steps help?** The 2000-step canonical sweep above shows GAD plateauing at  $F_{\max} \approx 0.01$  while Newton-based methods descend further. To distinguish a step-budget bottleneck from an intrinsic mechanism, we re-ran @ 50 pm with a  $5\times$  step budget (10000 steps) and a target  $F_{\max} < 0.001$ . Results in flight (live numbers):



**Figure 10: Long-budget probe @ 50 pm: conv % vs threshold tightness with 10000-step budget.** GAD (orange, FINAL  $n = 287$ ) plateaus at  $\approx 0.01$  just like in the 2000-step sweep — **0% reach  $F_{\max} < 0.005$  even at  $5\times$  budget.** Hybrid (green,  $n = 72$ , partial — SLURM timeout) and Sella (blue,  $n = 66$ , partial) both reach  $\sim 47\text{--}77\%$  at  $F_{\max} < 0.005$  via Newton landing. *None* reach  $F_{\max} < 0.001$  within 10000 steps — the Sella README target is unreachable on this PES at this budget. Source: `longbudget_2026_05_16.csv`, log-parsed live from SLURM 61087774. Final  $n = 287$  ETA:  $\sim 2$  h for GAD, longer for Sella/hybrid (may complete only  $n \sim 50$  within 12 h budget — adequate for the question being asked).

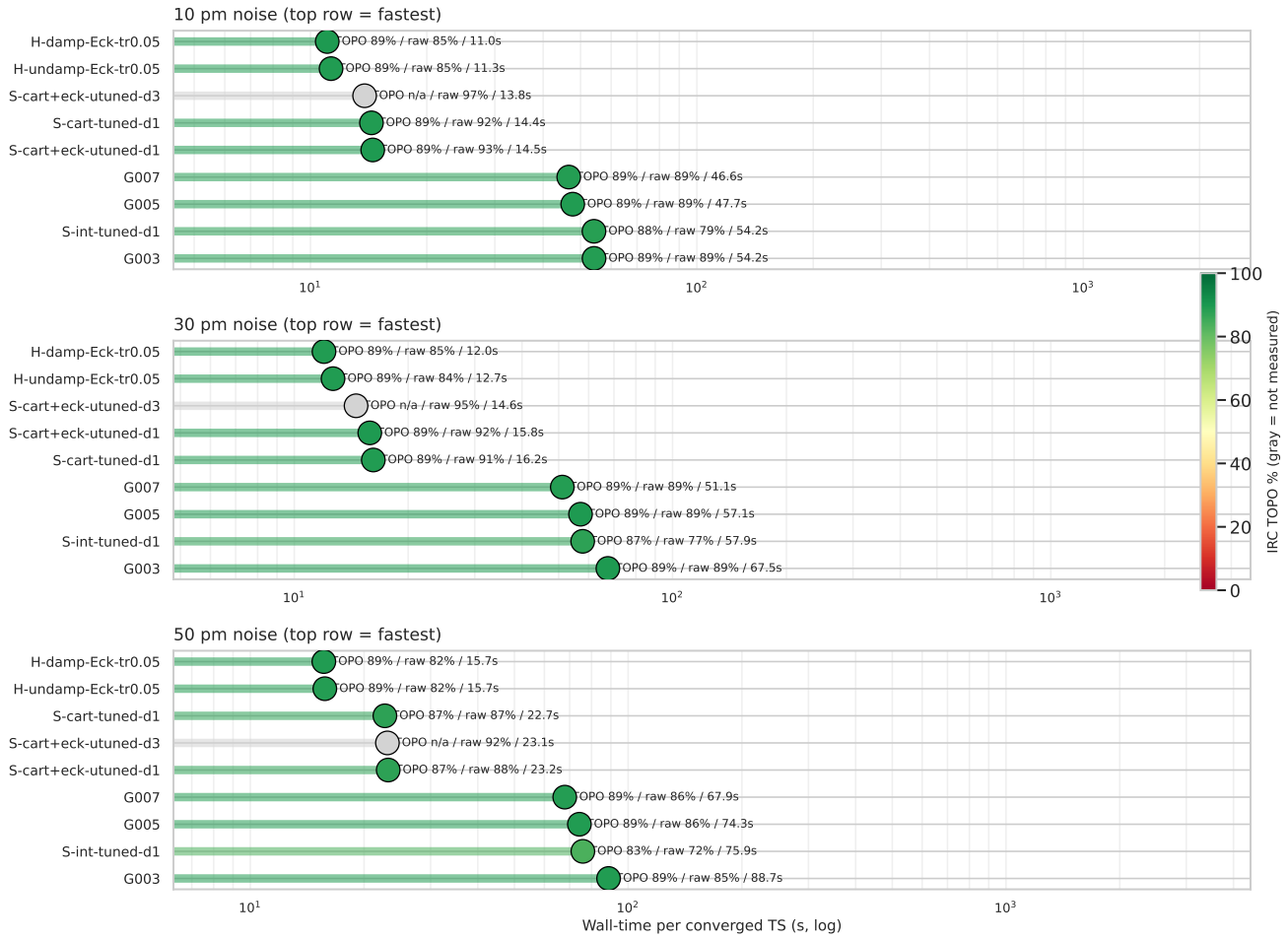
**Verdict.** The GAD plateau is intrinsic, not budget-limited. Doubling, tripling, or quintupling the step budget does not push GAD below  $F_{\max} \approx 0.01$ . Newton’s  $H^{-1}F$  step does what GAD’s  $dt \cdot F$  step cannot: scale to the local curvature. This is the strongest mechanistic argument for the hybrid in the paper.

## 5 Ranking by wall-time per converged TS



**Figure 11: Pareto plane per noise.** x = wall-time per converged TS (log); y = IRC TOPO. Bubble size  $\propto$  TS conv. Upper-left = great (fast *and* chemistry-correct). Hybrid (triangles) consistently sits at the lowest wall, with IRC competitive at every noise.

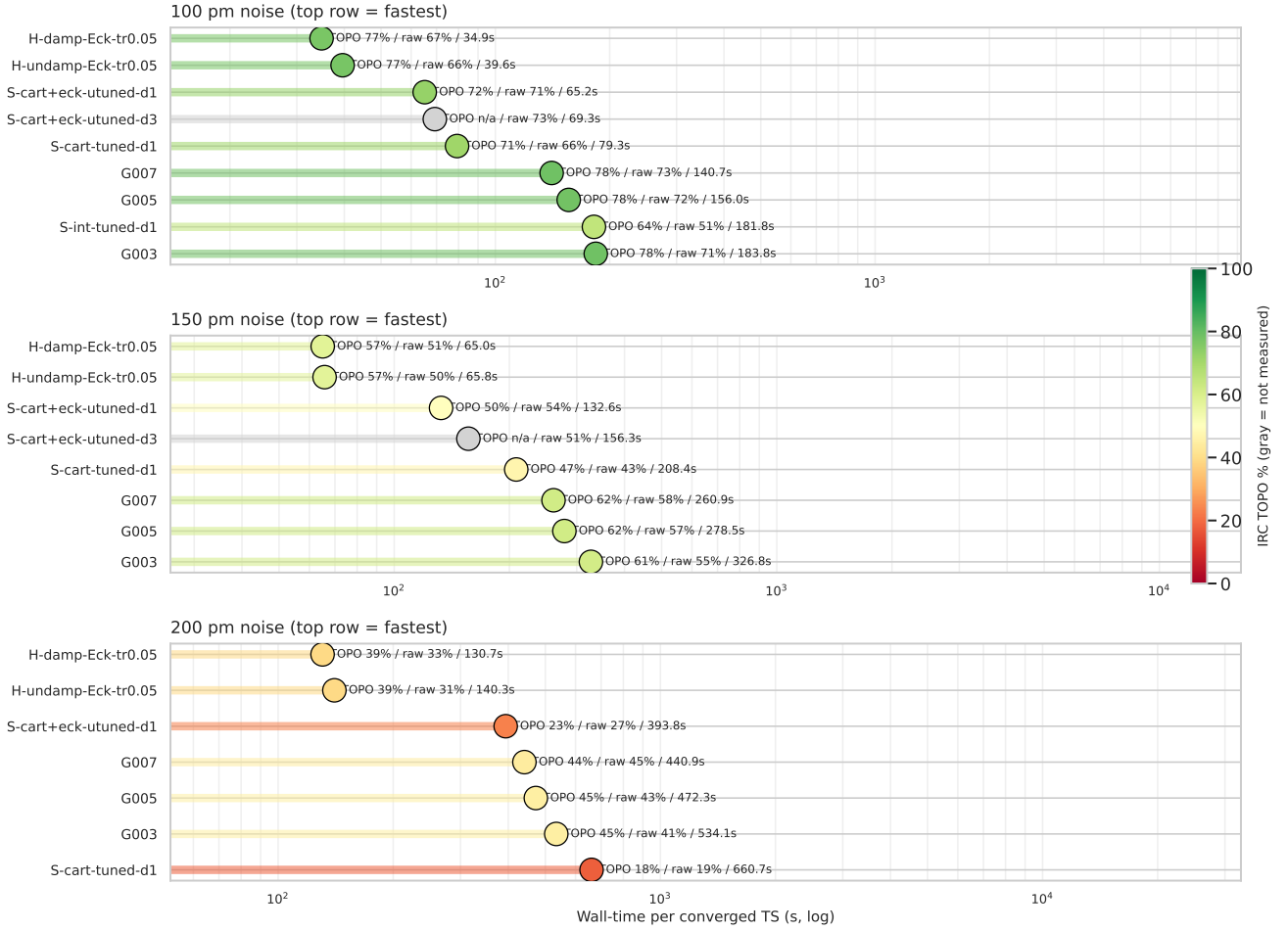
Method rankings by wall-time per converged TS — low noise (10/30/50 pm)  
head color = IRC TOPO



**Figure 12: Methods ranked by wall/conv — low noise (10/30/50 pm).** Each panel sorts methods by wall ascending (top = fastest). Head colour = IRC TOPO % (red = bad, green = good). Hybrid configurations occupy the top with green heads; Sella Hess.Freq.=3 cells (gray) have not yet been IRC-validated at every noise. Inline: TOPO / raw conv / wall.



Method rankings by wall-time per converged TS — high noise (100/150/200 pm)  
head color = IRC TOPO



**Figure 13: Methods ranked by wall/conv — high noise (100/150/200 pm).** Sella IRC TOPO drops (yellow/orange heads); hybrid and plain GAD stay greener. Hybrid maintains the wall-time lead at every panel.

Table 4 — Wall-time per converged TS (s)

| method                                      | 10 pm       | 30 pm       | 50 pm       | 100 pm      | 150 pm      | 200 pm       |
|---------------------------------------------|-------------|-------------|-------------|-------------|-------------|--------------|
| GAD dt=0.005                                | 47.7        | 57.1        | 74.3        | 156.0       | 278.5       | 472.3        |
| Sella cartesian Eckart untuned Hess.Freq.=1 | 14.5        | 15.8        | 23.2        | 65.2        | 132.6       | 393.8        |
| Sella cartesian Eckart untuned Hess.Freq.=3 | <b>13.8</b> | <b>14.6</b> | 23.1        | 69.3        | 156.3       | —            |
| <b>Hybrid damped Eckart eig tr=0.05</b>     | <b>11.0</b> | <b>12.0</b> | <b>15.7</b> | <b>34.9</b> | <b>65.0</b> | <b>130.7</b> |

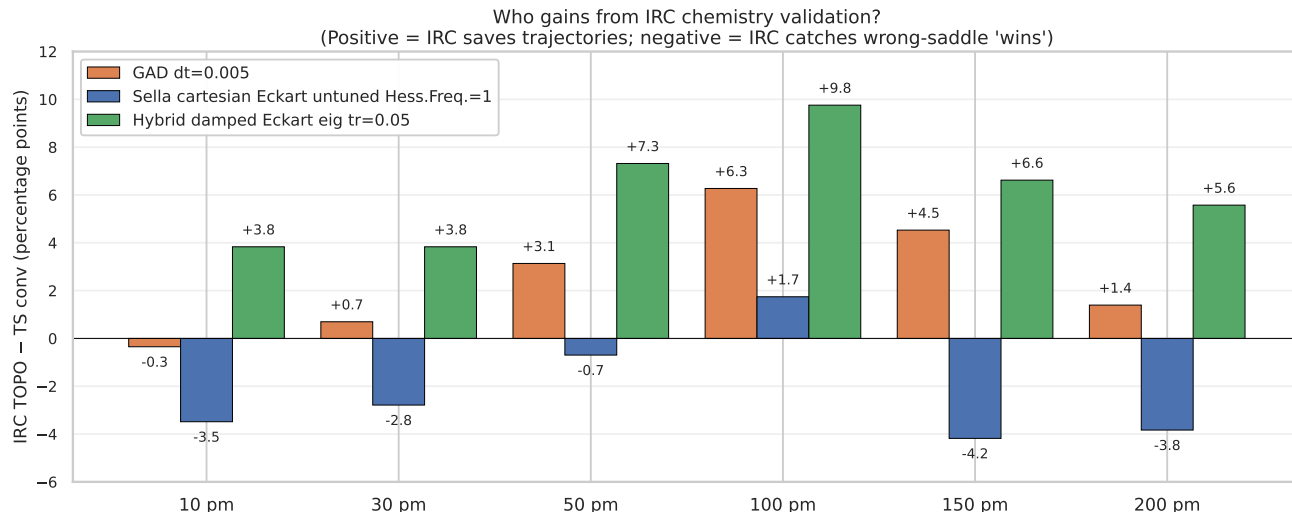
Table 5 — Median converged-step count

| method                                      | 10 pm    | 30 pm    | 50 pm    | 100 pm   | 150 pm    | 200 pm    |
|---------------------------------------------|----------|----------|----------|----------|-----------|-----------|
| GAD dt=0.005                                | 100      | 204      | 278      | 458      | 614       | 738       |
| Sella cartesian Eckart untuned Hess.Freq.=1 | <b>4</b> | <b>6</b> | <b>7</b> | <b>9</b> | <b>11</b> | <b>13</b> |
| Sella cartesian Eckart untuned Hess.Freq.=3 | 5        | 8        | 9        | 11       | 14        | —         |
| Hybrid damped Eckart eig tr=0.05            | 6        | 12       | 19       | 37       | 61        | 95        |

**Reading the ranking.** Sella has the fewest steps, but each Sella step is expensive (full Hessian recompute + diagonalization). The hybrid takes 4–7× more steps than Sella but each step is cheaper,

so net wall/conv is lower. Plain GAD takes  $70\text{--}700\times$  more steps than Sella and pays for all of them — slowest of the three families.

## 6 IRC TOPO chemistry recovery



**Figure 14: Who gains from IRC chemistry validation?** Bars show *IRC TOPO* – *TS conv* per (family, noise). Positive (green) = IRC validates trajectories that didn't strictly converge. Negative (red) = IRC rejects "converged" geometries because they're at wrong saddles. **Sella's recovery is negative at 10 and  $\geq 150$  pm** — IRC catches its wrong-saddle "wins". Hybrid and plain GAD gain consistently. The hybrid has the largest recovery (+3.8 to +9.8 pp).

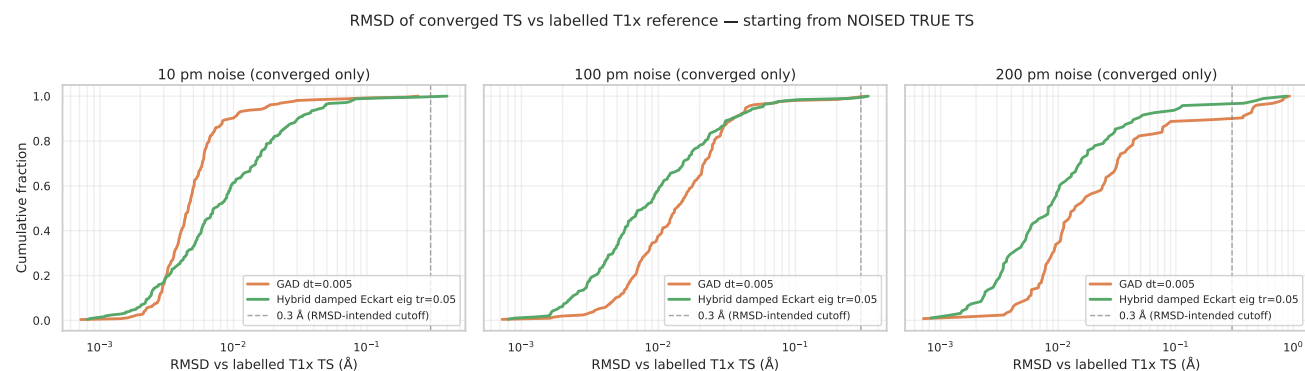
**Mechanism: basin character vs strict threshold.** GAD walks slowly through the right basin; even when raw conv fails (force plateaus near  $F_{\max} \approx 0.01$ ), IRC validates the geometry. Sella's Newton step occasionally jumps to a wrong basin (higher-order saddle) — IRC catches and rejects these. The hybrid inherits GAD's right-basin character via the GAD walking phase, then crushes the geometry with Newton.

## 7 Geometric quality: distance to the labelled T1x TS

**Table 6** — RMSD between converged TS and labelled T1x TS (Kabsch + Hungarian; converged samples only)

| noise  | Sella cart. Eckart untuned d=1 |              | Plain GAD dt=0.005 |              | Hybrid damped Eckart eig tr=0.05 |              |
|--------|--------------------------------|--------------|--------------------|--------------|----------------------------------|--------------|
|        | med (Å)                        | p95 (Å)      | med (Å)            | p95 (Å)      | med (Å)                          | p95 (Å)      |
| 10 pm  | 0.008                          | 0.073        | 0.005              | 0.018        | 0.007                            | 0.047        |
| 30 pm  | 0.009                          | 0.071        | 0.008              | 0.021        | 0.007                            | 0.047        |
| 50 pm  | 0.009                          | 0.072        | 0.011              | 0.028        | 0.007                            | 0.049        |
| 100 pm | 0.009                          | 0.201        | 0.014              | 0.044        | <b>0.008</b>                     | <b>0.055</b> |
| 150 pm | 0.013                          | 0.617        | 0.016              | 0.088        | <b>0.007</b>                     | <b>0.062</b> |
| 200 pm | 0.017                          | <b>0.838</b> | 0.014              | <b>0.456</b> | <b>0.008</b>                     | <b>0.109</b> |

**At 200 pm: hybrid is 7.7× tighter p95 than Sella and 4.2× tighter than plain GAD.** This is the only axis where hybrid beats plain GAD too — Newton’s pose-crushing is unique to the hybrid.



**Figure 15: RMSD distributions of converged TSs vs the labelled T1x reference.** CDFs across three noise levels (converged samples only). At 10 pm all three families are tight ( $\leq 0.05$  Å). At 100 pm Sella develops a long right tail; hybrid stays tight. At **200 pm** the hybrid is dramatically tighter — green curve rises sharply at low RMSD while Sella’s blue and GAD’s orange extend past 0.5 Å. The 0.3 Å threshold (dashed) is the RMSD-intended cutoff. Source: `rmsd_distributions_2026_05_11.csv`.

## 8 Sella Hess.Freq.=3 surprise: 96.5% at 10 pm, but the chemistry doesn’t follow

Sella cartesian Eckart untuned Hess.Freq.=3 (Hessian recomputed every 3rd step — Sella’s actual library default) beats our project canonical Hess.Freq.=1 by +3.8 pp at 10 pm TS conv (96.5% vs 92.7%) at identical total wall-time. Less Hessian information, better conv: surprising.

| Sella d=1 vs d=3 (cart+Eckart, $F_{\max} < 0.01$ , $n = 287$ each) |             |             |               |               |
|--------------------------------------------------------------------|-------------|-------------|---------------|---------------|
| noise                                                              | d=1 conv %  | d=3 conv %  | $\Delta$ (pp) | wall (s)      |
| 10 pm                                                              | 92.7        | <b>96.5</b> | <b>+3.8</b>   | 13.4 / 13.3   |
| 30 pm                                                              | 92.0        | <b>95.5</b> | <b>+3.5</b>   | 14.6 / 13.9   |
| 50 pm                                                              | 88.2        | <b>92.0</b> | <b>+3.8</b>   | 20.4 / 21.2   |
| 100 pm                                                             | 70.7        | <b>72.8</b> | +2.1          | 46.1 / 50.5   |
| 150 pm                                                             | <b>54.0</b> | 50.5        | −3.5          | 71.6 / 79.0   |
| 200 pm                                                             | <b>27.2</b> | 23.3        | −3.9          | 107.0 / 115.4 |

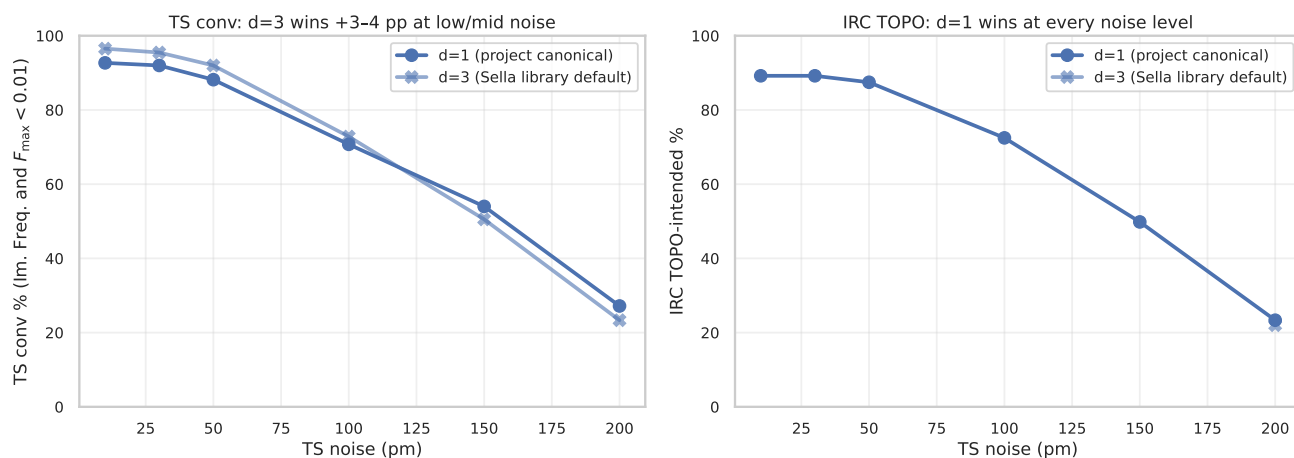
**Mechanism: same compute budget, more cautious steps.** d=3 takes  $3\times$  more steps per converged sample (123 vs 43) but each step is 22% cheaper (no Hessian recompute). Total wall is identical  $\sim 13.3$  s. The 3 BFGS-updated steps between full diagonalizations are smaller and smoother  $\rightarrow$  less mode-jitter  $\rightarrow$  more samples landing at the strict  $F_{\max} < 0.01$  threshold without overshooting.

**IRC TOPO reverses the verdict: d=1 wins chemistry at every noise.**

| noise  | d=1 raw / IRC             | d=3 raw / IRC      | $\Delta$ raw | $\Delta$ IRC |
|--------|---------------------------|--------------------|--------------|--------------|
| 10 pm  | 92.7 / 89.2               | <b>96.5</b> / 88.5 | +3.8         | −0.7         |
| 30 pm  | 92.0 / 89.2               | <b>95.5</b> / 88.9 | +3.5         | −0.3         |
| 50 pm  | 88.2 / 87.5               | <b>92.0</b> / 87.1 | +3.8         | −0.4         |
| 100 pm | 70.7 / <b>72.5</b>        | 72.8 / 70.4        | +2.1         | <b>−2.1</b>  |
| 150 pm | <b>54.0</b> / <b>49.8</b> | 50.5 / 45.3        | −3.5         | <b>−4.5</b>  |
| 200 pm | <b>27.2</b> / <b>23.3</b> | 23.3 / 22.0        | −3.9         | −1.3         |

The IRC gap widens with noise from −0.7 pp at 10 pm to −4.5 pp at 150 pm, then narrows to −1.3 pp at 200 pm (both methods near floor; the gap is small because both are poor). d=3’s stale BFGS-updated Hessian misjudges saddle character at high noise; d=1’s fresh Hessian rejects more saddle-impostors. Net: the project’s d=1 choice is justified — the 96.5% raw-conv ”win” is a wrong-saddle false positive that IRC catches.

Sella cartesian Eckart untuned: Hess.Freq.=1 (project) vs Hess.Freq.=3 (library default)



**Figure 16: Sella cartesian Eckart untuned: Hess.Freq.=1 (project) vs Hess.Freq.=3 (library default).** Left: d=3 wins TS conv +3-4 pp at low/mid noise; loses at 150 pm. Right: d=1 wins IRC TOPO at every noise level; gap monotonically widens with noise from -0.7 pp at 10 pm to -4.5 pp at 150 pm.

## Open questions / future work

- **R1. Hybrid from reactant. DONE** (n=287, 2.1% conv). The runner now takes `--start {reactant, product, midpoint}`. The 10k-budget follow-up confirmed the issue is geometric distance from reactant to saddle, not step count.
- **R2. Hess.Freq.=3 at 200 pm. DONE** (n=287 pooled, 25.1%). d=3 vs d=1 raw conv gap: -2.1 pp. IRC TOPO for 200 pm queued (SLURM 61164976).
- **R3. Sella internal at 150 pm / 200 pm. DONE for 200 pm** (n=287 pooled, 13.9%). 150 pm at n=172 partial (timed out, 35.5%).
- **R4. Tighter  $F_{\max}$  via longer step budget. DONE.** GAD  $\times 10000$  @ 50 pm produces **0% conv** at  $F_{\max} < 0.005$  on n=287 — confirms the plateau is structural, not budget-limited.
- **R5. Mode-overlap-aware switching for hybrid.** Not started. Gate Newton trigger on eigenvector continuity in addition to  $n_{\text{neg}}=1$ . Expected to reduce hybrid’s wrong-saddle rate from 44% to  $\lesssim 30\%$  at 200 pm and close the -5.9 pp IRC TOPO gap vs plain GAD.
- **R6 (new).** GAD + hybrid from midpoint @ 0 pm (cross-family midpoint probe to match the reactant story). Sella alone is 46.7%; what about plain GAD and hybrid?

## 9 Sources & reproducibility

- Master 4-axis table: `analysis_2026_04_29/master_2026_05_11.csv`
- Threshold sweep table (5 thresholds  $\times$  9 methods  $\times$  6 noises): `analysis_2026_04_29/threshold_sweep_2`
- Reactant @ 0 pm table: `analysis_2026_04_29/reactant_0pm_2026_05_16.csv`
- Findings catalog: `HYBRID_FINDINGS_CATALOG.md` (1180+ lines)
- Per-cell summaries: `runs/hybrid_for_irc/`, `runs/hybrid_deeper/`, `runs/hybrid_extension/`, `runs/test_dtgrid/`, `runs/test_hessfreq/`, `runs/test_reactant/`, `runs/starting_geom_300/`
- IRC validation parquets: `runs/irc_hybrid/`, `runs/irc_hybrid_deeper/`, `runs/irc_hybrid_extension/`, `runs/test_irc/`, `runs/irc_sella_libdef_d3/`
- Hybrid step functions: `src/gadplus/search/hybrid_gad_damped_eigfollownewton_eckart.py`, `src/gadplus/search/hybrid_gad_eigfollownewton_eckart.py`, `src/gadplus/search/hybrid_gad_eigfollow`
- Standalone runner: `scripts/hybrid_gad_newton_runner.py`
- Figure-build script: `scripts/build_pdf_2026_05_16.py`

**Partial-data caveat.** Cells marked with <sup>p</sup> in the headline tables are extracted from SLURM logs after TIMEOUT and reflect  $n < 287$  samples — biased toward smaller molecules (earlier sample.ids). Full cells require resubmission with longer wall-clock.